

William Boswell

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Industry Experience: **Technology, Finance, Healthcare, Telecom, Manufacturing**

Summary

Senior Data Engineer with 8+ years architecting scalable data solutions within the Microsoft ecosystem, specializing in Azure Data Factory, Synapse Analytics, and dimensional modeling. Proven track record building enterprise data pipelines that process billions of events daily while driving measurable business outcomes through modern data lakehouse architectures. Expert in leading cross-functional teams and implementing metadata-driven orchestration frameworks that enhance data quality, reliability, and organizational decision-making.

Education

Bachelor's Degree in Computer Science | University of Texas at Austin | May 2014

Professional Experience

Senior Data Engineer | Teradata | San Diego, CA | Sep 2022 – Present

- Architected comprehensive data platform strategy leveraging Azure Synapse Analytics, Data Factory, and Databricks to modernize Teradata's enterprise data infrastructure, resulting in 38% performance improvement across multi-region deployments
- Designed metadata-driven ingestion pipelines using Python, PySpark, and containerized services that standardized data flows from SAP BW and cloud storage systems, reducing deployment complexity by 70%
- Built dimensional models and star schemas optimized for analytics workloads, enabling faster reporting capabilities and supporting multimillion-dollar client expansion efforts
- Automated data orchestration workflows through Terraform infrastructure-as-code, ensuring consistent provisioning and reducing manual configuration steps while improving scalability
- Implemented comprehensive monitoring dashboards with Datadog and Grafana to track ingestion SLAs, data freshness, and pipeline reliability, decreasing incident detection time from hours to minutes
- Led technical workshops and design sessions with enterprise clients including Samsung and Audi, delivering proof-of-concepts capable of processing billions of monthly events
- Mentored data engineering teams on Azure best practices, dimensional modeling techniques, and distributed system design, fostering continuous learning culture
- Created detailed architectural documentation, data flow diagrams, and operational runbooks adopted across the global Center of Expertise

Data Engineer | New Relic | San Francisco, CA | Jan 2018 – Aug 2022

- Engineered real-time data ingestion platform using Azure Data Factory, Spark, and Databricks to process multi-tenant telemetry workloads exceeding 2 billion events daily for New Relic's analytics infrastructure
- Developed complex dimensional models and columnar data structures optimized for high-cardinality event data, significantly improving query performance while reducing storage overhead
- Implemented automated data quality frameworks using dbt and validation logic, decreasing downstream

data defects by 60% and ensuring reliable analytics for customer-facing dashboards

- Built secure API integration patterns with OAuth 2 authentication for enterprise client data ingestion, designing scalable JSON flattening and normalization processes
- Designed CI/CD workflows using GitHub Actions and Terraform to automate ETL deployment, testing, and environment provisioning across development and production environments
- Created observability dashboards measuring SLA adherence, pipeline latency, and data completeness metrics that improved operational reliability and customer satisfaction
- Collaborated with cross-functional teams including business analysts and cloud administrators to deliver comprehensive data solutions supporting high-value enterprise accounts

Software Engineer | Black Box Intelligence | Dallas, TX | Sep 2014 – Jan 2018

- Built end-to-end data integration pipelines consolidating POS, workforce, and consumer datasets from hundreds of external sources using SQL Server and Python-based ETL frameworks
- Designed normalization and dimensional modeling logic that improved data accuracy and consistency across benchmarking analytics products, enabling reliable business intelligence reporting
- Optimized SQL queries, indexing strategies, and stored procedures within relational database systems, achieving 55% reduction in batch processing runtime
- Implemented secure file transfer workflows and automated data ingestion processes, replacing manual operations and reducing nightly load failures by 80%
- Developed anomaly detection and data quality monitoring to proactively identify issues before impacting customer dashboards and analytics workflows
- Created aggregated fact and dimension tables supporting BI team requirements, enabling faster drill-down analytics and improved reporting capabilities

Software Engineering Intern | Microsoft | Austin, TX | Jun 2014 – Aug 2014

- Developed analytics tool improvements for cloud reliability evaluation, optimizing data ingestion and processing routines through targeted refactoring of internal Microsoft systems

Certifications

- Google Cloud Professional Data Engineer
- Databricks Certified Data Engineer Associate
- AWS Certified Solutions Architect - Associate

Skills

Cloud Platforms: Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure SQL Database, ADLS, Microsoft Fabric, AWS, S3

Programming Languages: Python, PySpark, SQL, T-SQL, JavaScript, PowerShell, DAX

Data Technologies: Apache Spark, Delta Lake, Dimensional Modeling, Star Schema, ETL Pipelines, Data Orchestration, Metadata-Driven Architecture

Databases: SQL Server, Azure SQL Database, PostgreSQL, NoSQL, Data Warehousing, Relational Databases

DevOps and Infrastructure: Terraform, Infrastructure as Code, Docker, GitHub Actions, CI/CD, Azure

DevOps, Containerization

Monitoring and Integration: Datadog, Grafana, OAuth 2, API Integration, JSON Processing, Data Quality Validation, dbt